

DULIPUDI LAASHMITH SANJAY

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SUMMARY

Electronics and Communication Engineering graduate with a strong foundation in Digital Electronics, Analog Electronics, CMOS VLSI, and Semiconductor Devices. Hands-on experience in digital system design, circuit design, simulation, Verilog HDL, and EDA tools. Aspiring to build a career as an RTL Design Engineer or Physical Design Engineer, with strong analytical thinking, problem-solving, adaptability, and collaboration skills.

EDUCATION

B.Tech - Electronics and Communication Engineering 2022 – 2026 | Guntur, India
Vasireddy Venkatadri Institute of Technology
CGPA: 8.35
Relevant Coursework: Digital Electronics, Analog Electronics, VLSI Design, Digital IC Design, Semiconductor Devices.

WORK EXPERIENCE

VLSI Design Intern 2025/05 – 2025/06 | Vijayawada
ChipIN Centre – C2S Programme, MeitY (Govt. of India)

- Designed and verified analog circuits including CMOS inverter, current mirrors, SRAM cell, and a two-stage CMOS operational amplifier, covering schematic design, simulation, custom layout, and post-layout verification.
- Developed Verilog RTL for combinational and sequential digital circuits, including adders, multipliers, counters, and finite state machines (FSMs), and verified functionality using simulation and waveform analysis.
- Performed RTL synthesis, timing analysis, floorplanning, placement, clock tree synthesis (CTS), and routing as part of the digital physical design flow using industry-standard EDA tools.

PROJECTS

64-Tap FIR Filter (RTL Design, Synthesis & Verification) 2025/05 – 2025/06

- Designed and implemented a parameterizable 64-tap FIR filter in Verilog HDL using custom multiplier and adder modules to develop a modular and scalable RTL architecture.
- Performed RTL synthesis by applying timing constraints and analyzing timing and area reports to optimize design performance and resource utilization.
- Developed a self-checking Verilog testbench and verified functional correctness through simulation and waveform analysis across multiple test cases.
- Validated the complete FIR filter design from RTL development to functional verification, ensuring reliable operation across multiple test scenarios.

Two-Stage CMOS Operational Amplifier 2025/05 – 2025/06

- Designed and implemented a two-stage CMOS operational amplifier with a differential input stage, current mirror load, and Miller compensation to achieve stable high-gain performance.
- Performed DC, AC, and transient simulations to evaluate gain, bandwidth, slew rate, phase margin, and overall circuit performance.
- Completed the full custom analog design flow from schematic capture to layout, followed by DRC/LVS verification and post-layout simulations to validate design accuracy.
- Optimized transistor sizing and circuit parameters to balance power consumption, stability, and gain-bandwidth performance while meeting design specifications.

SKILLS

Programming Languages — Verilog | C language | Python

EDA Tools — Cadence Virtuoso | Synopsys Design Compiler | Synopsys Verdi / ICC2 | Xilinx Vivado | MATLAB

Core Technical Skills — Digital Logic Design | RTL Design & Verification | CMOS Analog Circuit Design | Circuit Design & Simulation | Digital Physical Design Flow | Timing & Area Analysis

CERTIFICATES

- Digital Design for VLSI Chip Design - L&T EduTech [🔗](#)
- Digital Electronics - Infosys Springboard [🔗](#)
- VLSI Digital Design - Chip Design and Verilog Programming - Infosys Springboard [🔗](#)
- VLSI RTL Design & Verification Using Synopsys Tools - VLSIGuru Training Institute [🔗](#)
- The Joy of Computing Using Python - NPTEL